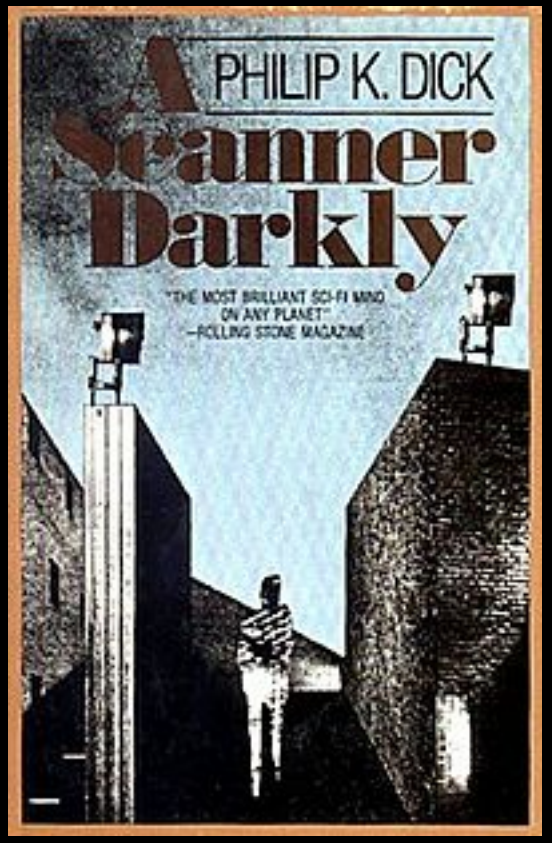
CS 0007: Introduction to Java Lecture 6

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Even more API for your first Object

SCANNERS

Scanner


PHILIP K. DICK
A Scanner Darkly
"THE MOST BRILLIANT SCI-FI MIND ON ANY PLANET"
—ROLLING STONE MAGAZINE

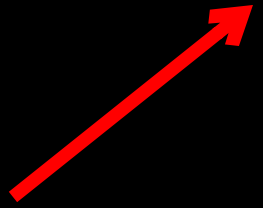
What is a Scanner?

- Retrieves input
- Can take input from a file, an input stream (console, network, etc.), or from a String
- Can read all primitives (except `chars`), and can read Strings
- An object

Declaring Objects

Type name = expression;

```
Scanner name = new Scanner();
```



We will fill this in with something,
but what exactly goes here?

Declaring Objects

- Any declaration of objects requires the keyword “**new**”
- The method that follows the keyword are called constructors
- Where can I look for them?

A Special `InputStream`

`System.in`

- The standard way to retrieve input (via the console)
- A “special” variable.
- Is this static or non-static?

```
Scanner consoleInput =
        new Scanner(System.in);
```

What can I do with it?

- I have a **`Scanner`** that can read input from the user via the console.
- How do I actually use it?

Using Scanner

```
public class AddingMachine
{
   public static void main(String[] args)
   {
         Scanner scan = new Scanner(System.in);
         System.out.println("Please enter a number:");
         double firstNum = scan.nextDouble();
         System.out.println("Please enter a second
   number:");
         double secondNum = scan.nextDouble();

         double sum = firstNum + secondNum;
         System.out.println("The sum of " + firstNum +
                " and " + secondNum + " equals " + sum);
   }//end method main
}//End class AddingMachine
```

Using Scanner

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   public static void main(String[] args)
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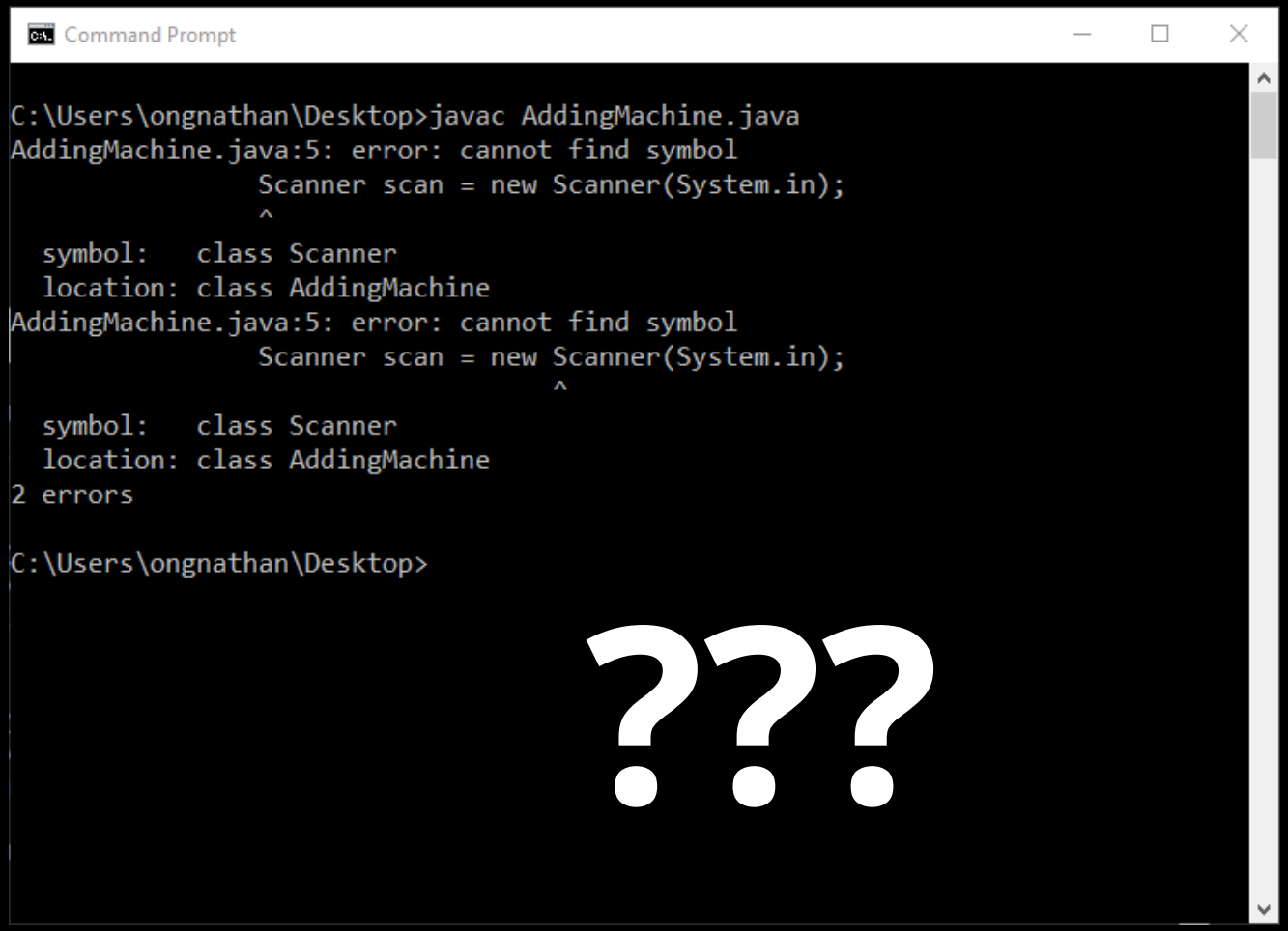
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Let’s Compile

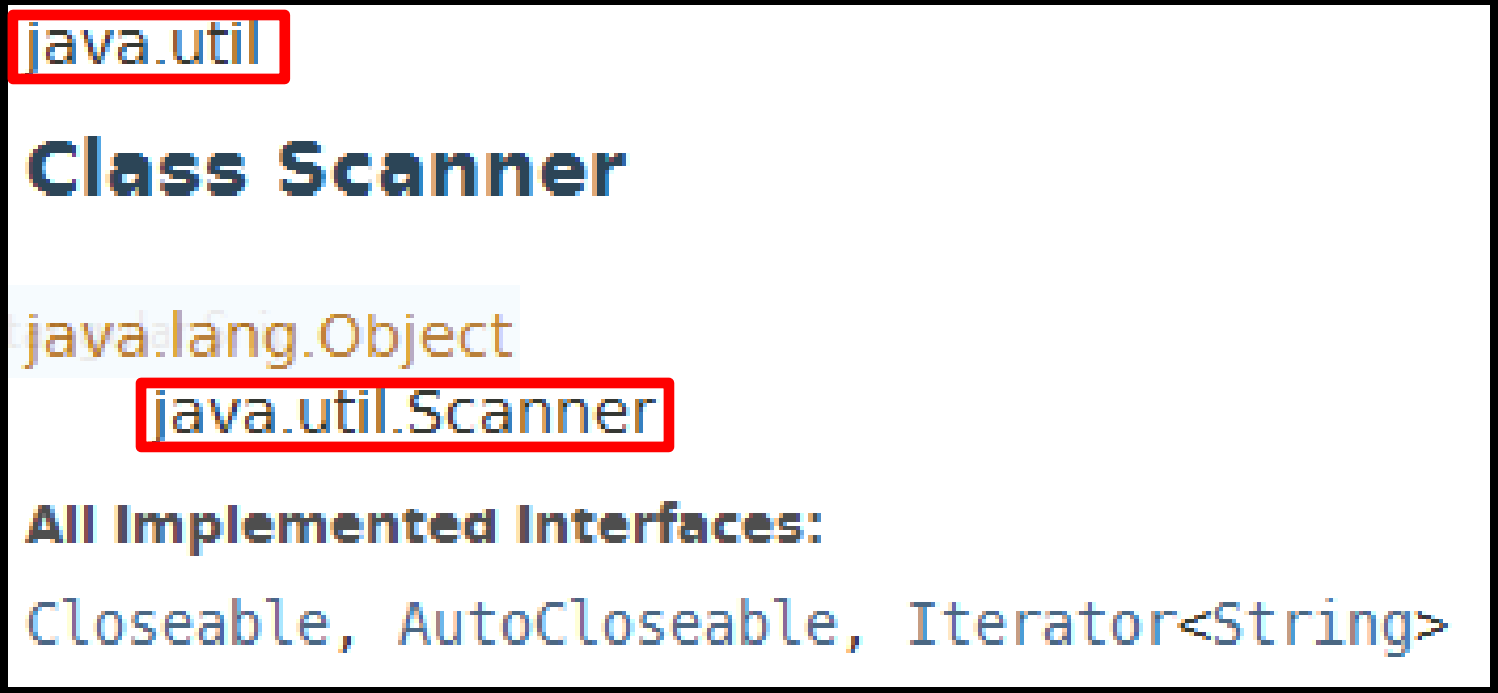


Error?!?

“Cannot find symbol”

- This indicates that the name does not refer to anything
- Based on the console output, “Scanner” does not exist
- Is there a way to figure out why?

From the API


java.util
Class Scanner
java.lang.Object
java.util.Scanner
All Implemented Interfaces:
Closeable, AutoCloseable, Iterator<String>

Error?!?

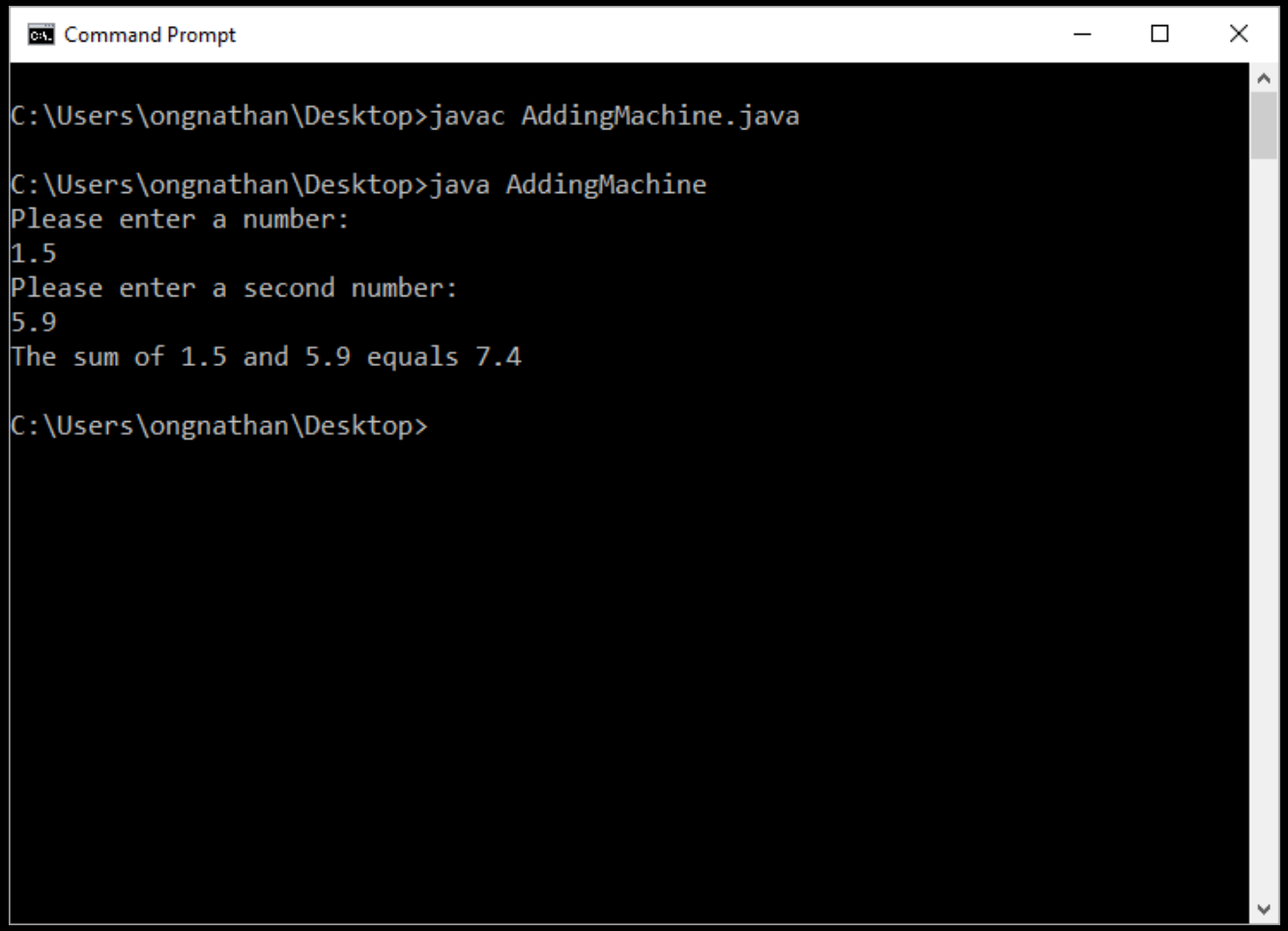
- Rather than being a part of “java.lang,” which is already included by default, it is part of a different package
- In order to include Scanner, we need to do an import

```java
import java.util.Scanner;

public class AddingMachine
{
   public static void main(String[] args)
   {
         Scanner scan = new Scanner(System.in);
         System.out.println("Please enter a number:");
         double firstNum = scan.nextDouble();
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   number:");
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         double sum = firstNum + secondNum;
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                " and " + secondNum + " equals " + sum);
   }//end method main
}//End class AddingMachine
```

Don’t Forget to Import



FUNCTIONS

Functions

- A function is similar to its mathematical counterpart
- $f(x) = x^2$, plug in 3, get 9
- Contains several more parts

Java Functions

- Takes in zero or more parameters, processes them in the function body, and returns a result
- Imagine going to BestBuy™ and telling them you want your computer fixed. You are telling them to run a fixing function, with your computer being a parameter. What you get back is your fixed computer.

You Already Have the Power!

- You already know how to call functions!

Static:
ClassName.functionName(<parameters>);

Non-static:
objectName.functionName(<parameters>);

- How do I make my own?

Function Components

Function Header

1. Visibility type (`public`/`protected`/`private`)
2. `static` (For now, required)
3. Return Type
4. functionName
5. Parentheses “()”
   - Parameters
   a) Type1 parameterName1
   b) Type2 parameterName2
   c) …

Function Body

6. Curly Brackets/Braces “{}”
   - `return` a value